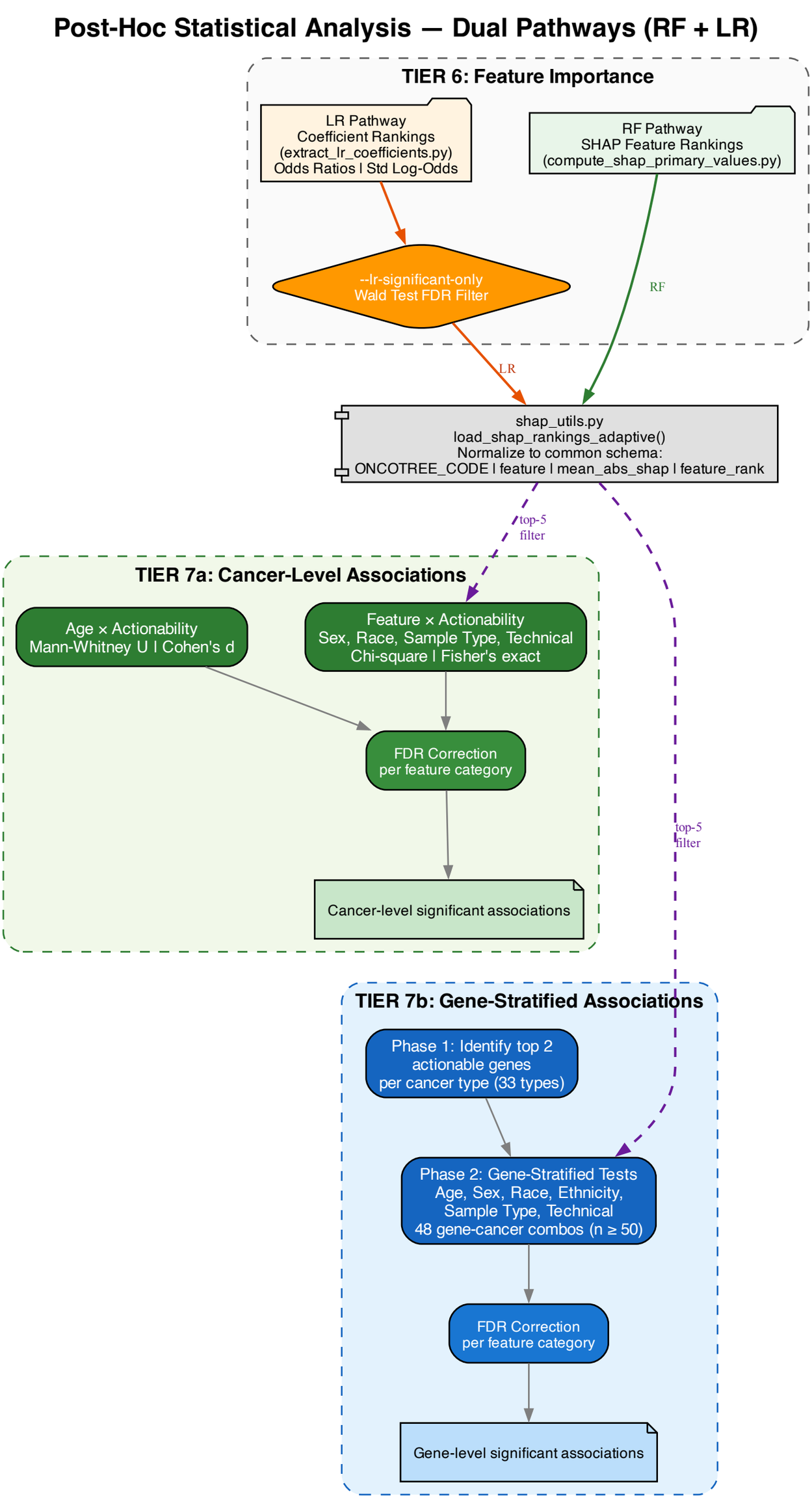

Post-Hoc Statistical Analysis — Dual Pathways (RF + LR)
TIER 6: Feature Importance
LR Pathway
Coefficient Rankings
(extract_lr_coefficients.py)
Odds Ratios | Std Log-Odds
RF Pathway
SHAP Feature Rankings
(compute_shap_primary_values.py)
--lr-significant-only
Wald Test FDR Filter
RF
LR
shap_utils.py
load_shap_rankings_adaptive()
Normalize to common schema:
ONCOTREE_CODE | feature | mean_abs_shap | feature_rank
top-5
filter
TIER 7a: Cancer-Level Associations
Age × Actionability
Mann-Whitney U | Cohen's d
Feature × Actionability
Sex, Race, Sample Type, Technical
Chi-square | Fisher's exact
FDR Correction
per feature category
Cancer-level significant associations
top-5
filter
TIER 7b: Gene-Stratified Associations
Phase 1: Identify top 2
actionable genes
per cancer type (33 types)
Phase 2: Gene-Stratified Tests
Age, Sex, Race, Ethnicity,
Sample Type, Technical
48 gene-cancer combos (n ≥ 50)
FDR Correction
per feature category
Gene-level significant associations